Target Detection Through Tree-Structured Encoding for Hyperspectral Images

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***Abstract*— Target detection aims to locate targets of interest within a specific scene. The traditional model-driven detectors based on signal processing have proved to be very effective. However, the detection performance of such traditional methods relies heavily on the model assumption, which is limited by the discrepancy with real hyperspectral images (HSIs) data. In this article, a target detection method through tree-structured encoding (TD-TSE) for HSIs is proposed. Instead of modeling the target and the background to extract valid features, we construct a binary tree based on the features of the data itself and segment the HSI to improve the separability of the target and the background. For the purpose of highlighting the target and suppressing the background, a novel measurement of separation, distance on tree, is calculated via binary encoding based on the constructed tree structure, and the detection output can be obtained according to such distance. To further reduce the generalization error resulting from random subsampling, the statistical average of the distances on multiple independent trees is estimated to improve the robustness of TD-TSE. The proposed method is not constrained by any model assumptions, which is fundamentally different from the most widely used hyperspectral target detectors in the field of signal processing. Moreover, the construction of binary trees without any labeled samples and the linear complexity of the proposed method make it highly practical for the hyperspectral data in real scenes. Extensive experiments on three benchmark HSI data sets demonstrate the effectiveness of the proposed TD-TSE for hyperspectral target detection.**

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***Index Terms*— Binary trees, encoding, hyperspectral image (HSI), target detection.**

I. Introduction

HYPERSPECTRAL remote sensing, with the abundant spectral information provided by the hyperspectral data, plays a critical role in a wide spectrum of applications [1], [2], [3]. Target detection focuses on separating the target of interest from background interference and is one of the most important application directions for hyperspectral images (HSIs) [4], [5], [6]. Based on the availability of the prior information of the target spectra, target detection plays a unique advantage in such tasks of finding specific objects [7]. Over the past few decades, hyperspectral target detection has been widely used and shown tremendous value in many fields of military and civilian [8], [9], [10].

Since the 1990s, some researchers have applied the signal processing theory to HSI processing and established a basic design framework for target detection in HSI [1]. The general processing flow of the classic hyperspectral target detection algorithms is to derive the decision functions via different mathematical models for the purpose of highlighting the target while suppressing the background [11], [12]. The test pixel represented by the vector is substituted into the decision function to generate the detection output [13], which is compared with a threshold to determine the existence of the target [14], [15]. The effectiveness of the detectors designed according to this design flow and its variations has been proven [1], [16]. So far, there are a large number of algorithms based on different model assumptions developed in the field of signal processing, and these can be divided into the following three categories [4], [17].

In the original space model, the detectors are based on the original spectral characteristics of HSI [18]. Euclidean distance (ED) performs matching via ED between the spectra of the target and the test pixel [5]. Spectral angle mapper (SAM) calculates the spectral angle error as the detection output [1].

The subspace projection model and the subspace projection transformation can be further divided into three types [19]: orthogonal projection, orthogonal subspace projection (OSP), and oblique projection. The projection operators are an important mathematical tool used in hyperspectral target detection and can realize the requirements of highlighting the target [20], suppressing the noise [21], and maximizing the

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signal-to-noise ratio (SNR) [22]. Representative OSP and its variants have shown to have a robust detection performance in practical applications [23]. The target-constrained interference minimized filter (TCIMF) designs a detection operator to constrain both the subspace of the target and the background [24], [25]. With the feature components of background being eliminated, the targets of interest can be detected with a lower false alarm rate (FAR).

In the probability statistics (white space) model, the decision function is derived based on the assumptions of conditional probability distribution under binary hypothesis testing [26]. The generalized likelihood ratio test (GLRT) depending on two hypotheses for the target and the background is also a widely used detector. The GLRT achieves a high SNR while maintaining a constant false alarm rate (CFAR) [13], [27], [28]. Based on the Fisher linear discriminant criterion, constrained energy minimization (CEM) distinguishes the target from the background by extracting signals in a specific direction while attenuating the signal interference in other directions [29].

In addition to the above three categories, there are other detectors that are designed based on mixed models. The adaptive coherence/cosine estimator (ACE) and adaptive matched filter (AMF) use the properties of the correlation or covariance matrix of the image under test [30], [31]; they are the two adaptive versions of GLRT [32]. The derivation of the decision functions is based on both the statistical theory and subspace projection. The aforementioned classic hyperspectral target detection methods are based on a mature theoretical system in the field of signal processing [1]. For the essential problem of highlighting the target and suppressing the background, different model assumptions are made on the HSI data to extract valid features. After obtaining the estimation of target and background-related information, the targets can be separated from the background [5].

Under reasonable assumptions, the above traditional algorithms perform well and exhibit robustness [33], [34]. However, there is an unavoidable bottleneck of such methods based on signal processing. If the description of the model for data in real scenes is not accurate enough [35], that is, the model does not match the real HSI data distribution well, the detection performance will decrease significantly [36]. In practice, hyperspectral sensors image on areas that contain multiple types of land cover, resulting in high complexity for both the target and the background. The traditional methods are theoretically constrained by the model assumptions, and the corresponding detectors cannot fully exploit effective features to accurately and reasonably estimate the target and background information, which directly affects the subsequent detection results [37].

Due to the limitations of the aforementioned traditional-model-driven methods based on signal processing, the introduction of machine learning has injected new vitality into the field of hyperspectral target detection [38], [39], [40]. A large number of machine-learning-based algorithms that emerged in recent years have achieved excellent detection performance and have unique advantages in many cases [41], [42], [43]. Representation-based detectors, which are an important migration in the field of machine learning, have become an active research area of hyperspectral target detection in recent years [44], [45]. According to the compressed sensing theory, the sparse-representation-based detector (SRD) aims to build both the target and background dictionaries to represent the test pixels in HSIs. The corresponding representation coefficients have only a few elements that are nonzero, that is, sparse. The recovered sparse representation is implemented for the detection output [46], [47]. For the test pixel, the combined sparse and collaborative-representation-based detector (CSCR) preserves the sparsity of the target, and the background components are collaboratively represented by its adjacent pixels [12]. The binary-class collaborative-representation-based detector (BCRD) adopts a background dictionary purification process for separating the target with more accurate background estimation [48]. Xu *et al.* [49] proposed a tensor robust principal component analysis (RPCA)-based HSI reconstruction and anomaly detection method. In this method, the background and anomaly parts are separated effectively by the novel Mahalanobis distance regularization and the detection accuracy is improved greatly. A large number of proposed representation-based methods and their derivative versions are successfully applied in hyperspectral target detection [50], [51]. However, most of these methods need to construct the target or background dictionary, which has high requirements for the prior spectral information. Moreover, some of them adopt the dual-window detection mode, and it is necessary to set appropriate sizes of dual window according to the spatial information of target to achieve the desired detection performance. Given limited prior information, such detectors are restricted to some extent in practical applications.

As a classic structure in machine learning, binary tree has been widely used for segmenting high-dimensional data, which is the core operation of classification from the perspective of mathematics [52]. To determine whether the test pixel is target or background, hyperspectral target detection is essentially a binary classifier [48], [53]. Although the tree structure is very suitable for processing classification problems [54], the research on such method migration for hyperspectral target detection is limited. Compared with classification, the object of interest for target detection exists in the form of fewer and different data points in HSIs, which requires a higher evaluation accuracy [55]. Moreover, after separating various types of land cover, most tree-based classifiers do not reasonably measure the separation of different classes for the purpose of highlighting the target while suppressing the background, resulting in the inability to output the detection results [56], [57]. Because of this, in this study, we adopt a novel measurement of separation, distance on tree, to design the output of the detector, so that the tree structure can be fully exploited in target detection for HSIs and maximum advantage. In particular, considering the correspondence between the binary tree and binary encoding, 0–1 codes are used to implement the calculation of distance on tree.

Specifically, the target detector through tree-structured encoding (TD-TSE) proposed in this article constructs the binary tree by subsampling and aims to improve the separability of the target and the background. Starting from the root, a test pixel will eventually fall into a leaf node, and the

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0–1 binary encoding is performed accordingly. For the given prior knowledge of the target spectrum, the distance on tree between the test pixel and the target is calculated with their binary codes. For each pixel in the HSI, the detection output can be obtained based on the distance on tree. The novelty and contributions of the proposed TD-TSE can be summarized as follows.

1) Without being constrained by specific model assumptions, the construction of the binary tree depends entirely on the features of the HSI data itself and seeks to make the tree structure more adaptable to different scenes. The spectral characteristics of the target are essentially reflected in the difference between its value and that of the surrounding background at certain wavelengths. For the segmentation process based on the spectral characteristics of the HSI, the nodes on tree at different layers continuously divide the data into two subsets and gradually improve the separability of the target and the background.
2) The detection output of the proposed method is designed based on a novel measurement of separation, the distance on tree between the test pixel and the given target. Based on the constructed tree structure, the distance on tree can be skillfully calculated via binary encoding. The spectral difference between the target and the background is reflected in the numerical separability; as a result, they fall into distant leaf nodes and the distance on tree is defined by the shortest path length between two leaf nodes. Such graphical distance offers a reasonable measurement to quantify the spectral difference and is more in line with the objective of highlighting the target and suppressing the background for hyperspectral target detection.
3) Considering the generalization error caused by random subsampling, we calculate the statistical average of distances on multiple independent trees to make the proposed method more robust. For such error reduction through parallel measurements, although the ensemble learning theory is applied, unlike typical forest-based methods such as random forest and isolation forest, TD-TSE does not require the use of any labeled samples, not even a large number of trees. This is largely due to the designed measurement, distance on tree, calculated based on the graphical structure of the tree. In addition, the worst case requirements of memory and processing time increase linearly with the subsampling size and the number of trees, respectively, facilitating practical applications.

The rest of the article is organized as follows. Section II provides a detailed explanation of the proposed TD-TSE. Section III reports on the extensive experimental results and analyzes the performance of the detector. Finally, the conclusion is drawn in Section IV.

II. Target Detection Through Tree-Structured Encoding

The flowchart of the proposed TD-TSE is shown in Fig. 1. The algorithm input consists of a HSI in a specific scene and the given target spectrum. The processing can be divided into three parts: 1) implementation of segmentation using the binary tree constructed by subsampling to improve the separability of the target and the background; 2) encoding the samples according to the leaf nodes that they fall into and calculating the distance on tree between the test pixel and the target with their binary codes; and 3) constructing a specific number of trees by multiple subsampling and then outputting the detection result based on the statistical average of the distances. Detailed explanations of these three parts are given in parts A, B, and C of Section II, respectively.

A. Segmentation Based on Construction of the Binary Tree

A tree is a hierarchical data structure defined by branch relationships in computer science. The nodes are the basic constituent units of a tree and can be divided into "child" and "parent" according to the connection form. The root nodes have no parent, while the leaf nodes have no child. In addition to the root, each child can be divided into multiple disjoint subtrees. A binary tree is defined as having at most two subtrees per node [58], [59].

By applying the tree structure to a HSI, the data set can be continuously divided into two parts by the nodes on tree until it is indivisible and all the leaf nodes correspond to disjoint subsets. The unique structural characteristics of the tree enable the segmentation of HSIs, which is the core operation in hyperspectral classification from the mathematical perspective [60]. Considering the spectral difference between various classes of land cover, as a binary classification, it is reasonable for target detection to use the tree structure to improve the separability of the target and the background [61]. For HSI in a specific scene, the given target usually appears in the form of few and different data points [62]. The characteristic of the target sample is specifically manifested in its spectrum, which is different from that of the surrounding background [5]. More specifically, the target spectrum has larger or smaller values at all or some wavelengths. However, after the binary tree is implemented for image segmentation, only pixels with high spectral similarity or strong correlation exist in the same or close leaf nodes, which means that the target and the background are divided into segmented regions corresponding to distant leaf nodes. Accordingly, the objects of interest can be separated from the background effectively. Besides, due to the limitations resulting from the spatial resolution and the complexity of the land cover distribution, the target of interest often appears in the form of subpixels mixed with the background [63]. In this regard, the main difficulty of target detection for HSIs is that there is a certain spectral similarity between the subpixel target and the background, causing the decrease in separability [28]. Therefore, to divide the pixels of target and background into different subsets at the nodes of specific layers, we apply the number of bands in the original HSI to constrain the depth of the tree during construction. Without losing spectral information in any dimension, the subsequent tree-based segmentation ensures that the subpixel target and background can be separated at

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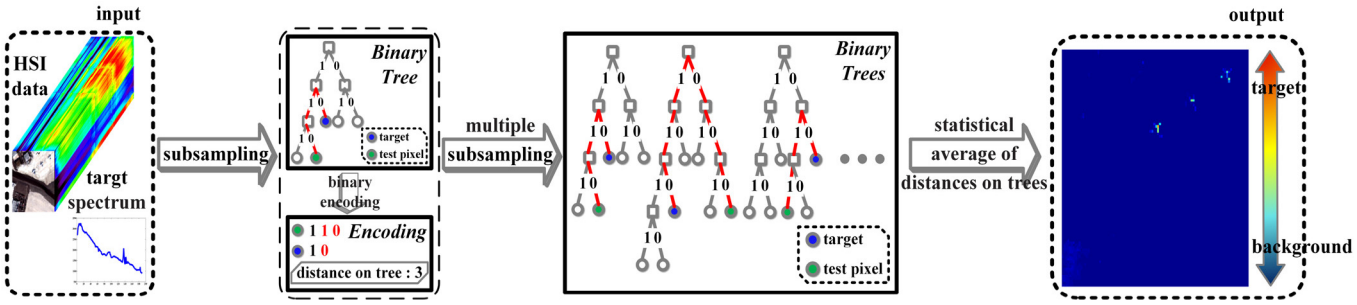


Fig. 1. Flowchart of the proposed TD-TSE for HSI.

less correlated spectral bands and eventually fall into two leaf nodes that are distant from each other.

Algorithm 1 get_one_tree($\mathbf{X}_{\text{sub}}$, $\text{depth}_{\text{limit}}$, $\text{depth}_{\text{current}}$, $\mathbf{dim}_{\text{use}}$)

Inputs: $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{L\times \text{tree}_{\text{size}}}$-a subset of the HSI, $\text{depth}_{\text{limit}}$-the depth limit of the tree, $\text{depth}_{\text{current}}$-the current depth of the tree, $\mathbf{dim}_{\text{use}} \in \mathbb{R}^{L\times 1}$-the band numbers corresponding to the sequential layers of the tree.
Output: tree-a binary tree.
Initialization: $\text{depth}_{\text{limit}} = L, \text{depth}_{\text{current}} = 0$, $\mathbf{dim}_{\text{use}}$ denotes integers from 1 to L after random shuffling.

```
X_remain = X_sub;
if depth_current < depth_limit and X_remain is separable then
dim_current = dim_use(depth_current + 1),
X_dim = X_remain (dim_current,:);
randomly select a segmentation point value_seg as the
node of the tree between the maximum and minimum
values of X_dim, tree · node = value_seg;
X_left = X_remain (:, find(X_dim <= value_seg));
X_right = X_remain (:, find(X_dim > value_seg));
depth_current = depth_current + 1;
tree · left_branch = get_one_tree(X_left, depth_limit,
depth_current, dim_use);
tree · right_branch = get_one_tree (X_right, depth_limit,
depth_current, dim_use);
end if
return tree
```

In summary, the tree structure used in the proposed detector provides a reliable segmentation strategy for separating the target and the background. For a given HSI data set $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$, $\mathbf{X} \in \mathbb{R}^{L\times N}$, L and N are the number of bands and pixels, respectively. We randomly select $\text{tree}_{\text{size}}$ pixels to form a subset $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{L\times \text{tree}_{\text{size}}}$, where $\mathbf{X}_{\text{sub}} = \left[\mathbf{x}_1, \ldots, \mathbf{x}_{\text{tree}_{\text{size}}}\right]$, $\mathbf{X}_{\text{sub}} \subseteq \mathbf{X}$, $\text{tree}_{\text{size}} < N$, $\text{tree}_{\text{size}}$ refers to the subsampling size. A binary tree is constructed by recursively dividing $\mathbf{X}_{\text{sub}}$ into two parts until the current depth of the tree reaches the limit or the subset at the node is inseparable. The specific steps of the construction of a binary tree are shown in Algorithm 1. Table I provides necessary explanations for the symbols that appear in the description of the algorithms.

For the final binary tree obtained at the end of the construction, the depth reaches the number of bands or the values on each leaf node are equal. For tree construction, the worst case indicates that the binary tree fully grows and each pixel $\mathbf{x}_i \in \mathbb{R}^{L\times 1}, i = 1, \ldots., \text{tree}_{\text{size}}$ in $\mathbf{X}_{\text{sub}}$ falls into $\text{tree}_{\text{size}}$ different leaf nodes. All $2 \times \text{tree}_{\text{size}} - 1$ nodes on the tree include $\text{tree}_{\text{size}}$ leaf nodes and $\text{tree}_{\text{size}} - 1$ nonleaf nodes in this case. The memory requirement is linearly related to the subsampling size $\text{tree}_{\text{size}}$, which makes the proposed TD-TSE applicable to large-scale hyperspectral data sets.

B. Binary Encoding for Distance on Tree

After the binary tree structure has been implemented for segmentation to separate the target of interest from the background interference, the focus is on designing the output of each test pixel to give a reasonable basis for determining whether it is a target or not. The crucial requirement for the detection output is that it should highlight the target while suppressing the background. To this end, the proposed TD-TSE uses a novel separation measurement called the distance on tree. Based on the tree topology, which reflects the graphical connection relationship between all the nodes, the distance on tree is defined by the shortest path length between two leaf nodes. As shown in Fig. 1, it is the number of branches marked in red passing from the test pixel to the target. The graphical distance is quantified using the concept of encoding. As there are different paths available from the root to different leaf nodes, it is natural to record the path by binary encoding and the distance on tree between leaf nodes can be measured with their 0–1 codes.

For the given target spectrum in the form of a vector, put it into the root node of the constructed binary tree. Different dimensions of the vector correspond to different layers of the tree; the target falls into either the left or right branch depending on whether its value in a certain dimension is less than or greater than the node; eventually, the target ends up in a leaf node. The same operation is performed on each pixel in the HSI and the path from the root to the leaf is obtained. The topology of the tree reflects the connection relationship between all the nodes, which results in the paths corresponding to different leaves being exclusive. As the paths can be recorded by binary encoding [64], the given target and all the pixels in the HSI have their own 0–1 codes whose highest bits denote the branch of the root node. For the codes indicating the target and test pixel, align the highest bits of them, start comparing, and find the first position that appears different. This position denotes the deepest node shared by the

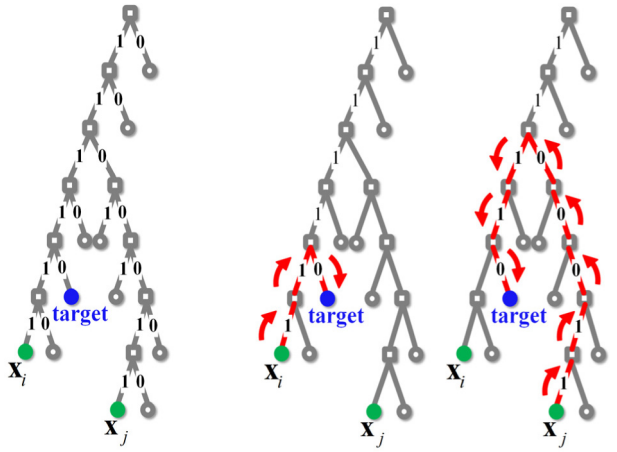
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TABLE I
INTRODUCTION TO SYMBOLS

Symbol	Symbolic Representation
$\mathbf{X}$	a hyperspectral image (HSI)
$\mathbf{X}_{sub}$	a subset of $\mathbf{X}$
L	the number of bands in $\mathbf{X}$
N	the number of pixels in $\mathbf{X}$
$\mathbf{x}_i$	the i-th test pixel in $\mathbf{X}$
target	the given target spectrum
tree	*tree* -a binary tree, which is a struct with three members. *tree.node* -the node of the tree *tree.left _branch* -the left branch of the tree *tree.right _branch* -the right branch of the tree
$\mathbf{tree}_{set}$	the collection of multiple binary trees, a set of *tree*
$tree_{size}$	the subsampling size
$tree_{num}$	the number of trees
$depth_{limit}$	the depth limit of *tree*
$depth_{current}$	the current depth of *tree*
$\mathbf{dim}_{use}$	the band numbers of different dimensions of the HSI corresponding to the sequential layers of *tree*
$dim_{current}$	the band number of a certain dimension of the HSI corresponding to the current layer of *tree*
$\mathbf{X}_{remain}$	the remaining dataset in the process of building *tree*
$\mathbf{X}_{dim}$	the $dim_{current}$-th band of $\mathbf{X}_{remain}$
$\mathbf{X}_{left}$	the data in $\mathbf{X}_{remain}$ divided into the left branch
$\mathbf{X}_{right}$	the data in $\mathbf{X}_{remain}$ divided into the right branch
$distance_{avg}$	the statistical average of the distances on multiple trees
result	the detection result of a test pixel
result	the detection results of the HSI, a set of *result*

paths of the target and the test pixel and can be used to measure the distance on tree between them; the shorter the distance, the higher the likelihood of the test pixel being judged as a target.

Assume that $\mathbf{x}_i$, $\mathbf{x}_j \in \mathbb{R}^{L\times 1}$ are two pixels in the given HSI data set $\mathbf{X}$, **target** $\in \mathbb{R}^{L\times 1}$ refers to the given target spectrum, and $\mathbf{x}_i$, $\mathbf{x}_j$, and **target** fall into different leaf nodes of the binary tree, as shown in Fig. 2(a). Encoding $\mathbf{x}_i$, $\mathbf{x}_j$, and **target** based



(a) (b)

Fig. 2. (a) Encoding based on the binary tree structure. (b) Distance on tree between the test pixel and the target.

on the tree structure, we have

$$\begin{aligned} \mathbf{code}_{\mathbf{x}_i}: &\quad 1\ 1\ 1\ 1\ 1\ 1 \\ \mathbf{code}_{\mathbf{x}_j}: &\quad 1\ 1\ 0\ 0\ 0\ 1\ 1 \\ \mathbf{code}_{\mathbf{target}}: &\quad 1\ 1\ 1\ 1\ 0\,. \end{aligned} \tag{1}$$

The highest bits of the 0–1 codes in (1) are aligned. Compare $\mathbf{code}_{\mathbf{x}_i}$ with $\mathbf{code}_{\mathbf{target}}$ and find the first position that appears different, marked with the horizontal bar above. The shortest path length from the leaf node of $\mathbf{x}_i$ to the leaf node of **target** is calculated as follows:

$$\begin{aligned} \mathbf{code}_{\mathbf{x}_i}: &\quad 1\ 1\ 1\ 1\ \bar{1} \leftarrow 1 \\ &\qquad\qquad\quad \downarrow \\ \mathbf{code}_{\mathbf{target}}: &\quad 1\ 1\ 1\ 1\ \bar{0} \end{aligned} \tag{2}$$

where $1 \rightarrow \bar{1} \rightarrow \bar{0}$ indicates the record of the path. So the distance on tree between $\mathbf{x}_i$ and **target** is 3. Similarly, for $\mathbf{x}_j$, the shortest path length can be obtained by

$$\begin{aligned} \mathbf{code}_{\mathbf{x}_j}: &\quad 1\ 1\ \bar{0} \leftarrow 0 \leftarrow 0 \leftarrow 1 \leftarrow 1 \\ &\qquad\quad \downarrow \\ \mathbf{code}_{\mathbf{target}}: &\quad 1\ 1\ \bar{1} \rightarrow 1 \rightarrow 0\,. \end{aligned} \tag{3}$$

The path record from $\mathbf{x}_j$ to **target** is $1 \rightarrow 1 \rightarrow 0 \rightarrow 0 \rightarrow \bar{0} \rightarrow \bar{1} \rightarrow 1 \rightarrow 0$, and so the distance on tree in this case is 8. Fig. 2(b) shows the branches passing from $\mathbf{x}_i$ and $\mathbf{x}_j$ to **target**. The minimum number of branches marked in red confirms the calculation results. According to the topology shown in Fig. 2, it can be seen that $\mathbf{x}_i$ is separated by the node in the deeper layer closer to **target**, while $\mathbf{x}_j$ is separated in the shallower layer farther from **target**, which results in $\mathbf{x}_i$ being closer to **target** than $\mathbf{x}_j$ is in the tree. Obviously, the overall spectral similarity between $\mathbf{x}_i$ and the given target is higher from the perspective of numerical separability, so $\mathbf{x}_i$ is more likely to be judged as a target. The degree of spectral similarity between the test pixel and the target is intuitively shown in the tree structure. Therefore, it is reasonable for the proposed TD-TSE method

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to give the possibility of the existence of the target based on the distance on tree.

Algorithm 2 TD-TSE ($\mathbf{X}$, **target**, $\text{tree}_{\text{size}}$, tree_{num})

Inputs: $\mathbf{X} \in \mathbb{R}^{L\times N}$-the input HSI, **target** $\in \mathbb{R}^{L\times 1}$-the given target spectrum, $\text{tree}_{\text{size}}$-the subsampling size, tree_{num}-the number of trees.
Output: **result** $\in \mathbb{R}^{1\times N}$-the detection results.
Initialization: $\text{depth}_{\text{limit}} = L$, $\text{depth}_{\text{current}} = 0$.
1: **for** $k = 1 : \text{tree}_{\text{num}}$ **do**
2: randomly select $\text{tree}_{\text{size}}$ pixels to form a subset $\mathbf{X}_{\text{sub}} \in \mathbb{R}^{L\times \text{tree}_{\text{size}}}$;
3: randomly shuffle integers from 1 to L to get $\mathbf{dim}_{\text{use}}$;
4: $tree$ = get_one_tree ($\mathbf{X}_{\text{sub}}$, $\text{depth}_{\text{limit}}$, $\text{depth}_{\text{current}}$, $\mathbf{dim}_{\text{use}}$);
5: $\mathbf{tree}_{\text{set}} = \mathbf{tree}_{\text{set}} \cup tree$;
6: **end for**
7: encode **target** based on $\mathbf{tree}_{\text{set}}$ to get tree_{num} sets of 0–1 codes;
8: **for** $i = 1 : N$ **do**
9: $\mathbf{x}_i \in \mathbb{R}^{L\times 1} \leftarrow i$-th test pixel in $\mathbf{X}$;
10: encode $\mathbf{x}_i$ based on $\mathbf{tree}_{\text{set}}$ to get tree_{num} sets of 0–1 codes;
11: measure tree_{num} sets of distances on trees between $\mathbf{x}_i$ and **target**;
12: calculate the statistical average of tree_{num} sets of distances to obtain $\text{distance}_{\text{avg}}$;
13: result = $\exp(-\text{distance}_{\text{avg}})$;
14: **result** = **result** $\cup$ result;
15: **end for**
16: **return result**

C. Reduction of Generalization Error Based on Multiple Binary Trees

The aforementioned binary tree used for segmentation is constructed by subsampling. For the given HSI, the number of bands is applied as the height limit of the tree to avoid the loss of spectral information. However, subsampling by random selection of pixels cannot preserve the complete spatial information. In this case, the generalization error will inevitably occur in the measurement of the distance on tree; this directly affects the detection result and the subsequent judgment regarding the existence of the target. Based on the theory of ensemble learning, such negative effects can be reduced by increasing the number of binary trees and by taking the arithmetic average using parallel measurements [65], [66]. According to this, we implement such strategy for the reduction of generalization error to improve the robustness of TD-TSE.

Putting the above considerations together, the execution procedures are integrated as follows. First, randomly select a certain number of pixels in the HSI multiple times to build independent binary trees. Second, for the given target spectrum, record the path from the root to the leaf node on all trees with binary encoding and obtain multiple sets of 0–1 codes. Perform the same operation on each pixel in the HSI and then we can get multiple distances from the pixel to the target corresponding to different trees. Finally, for each test pixel, calculate the statistical average of the distances. The output of the proposed detector is obtained based on the average distance. The overall process of the TD-TSE is shown in Algorithm 2.

The input of Algorithm 2 includes two important parameters; $\text{tree}_{\text{size}}$, tree_{num} indicate the subsampling size and the number of trees, respectively. The worst case linear relationship between the memory requirement and $\text{tree}_{\text{size}}$ has already been discussed in Section II-A. It can be deduced that the processing time is also linearly related to the number of trees tree_{num}. Theoretically, the proposed TD-TSE with linear complexity is suitable for processing HSI data sets with a large number of samples. In the subsequent experiments of TD-TSE, it should be noted that the settings of $\text{tree}_{\text{size}}$ and tree_{num} both affect the detection performance. In addition, the optimal parameter combinations of $\text{tree}_{\text{size}}$ and tree_{num} are different for different HSI data sets. The settings of the experimental parameters will be discussed in more detail in Section III.

III. Experimental Results and Analysis

Extensive experiments on three benchmark HSI data sets were made to demonstrate the effectiveness of the proposed method. Section III-A gives the detailed description of the experimental data sets and quantitative evaluation indicators. In Section III-B, the optimal settings of the main parameters for different HSI data sets are discussed. Section III-C deals with the comparisons of the performance between the proposed method and other classic and state-of-the-art target detection algorithms. All experiments are carried out using MATLAB R2014a on a Windows 7 computer with an Intel Core i3 CPU at 3.07 GHz and 4.00 GB RAM.

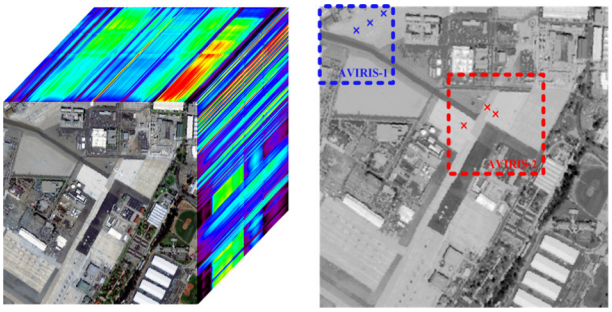
A. Experimental HSI Data Sets and Evaluation Indicators

Three subimages used in our experiments all come from publicly available benchmark data sets for hyperspectral target detection.

1) AVIRIS-1 Data Set: The first data set was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over San Diego. The original image includes 224 spectral channels in the wavelength range of 370–2510 nm. We reserved 189 bands in the experiments, leaving out the water absorption bands and low SNR bands [9]. The full image of the urban scene contains 400 × 400 pixels and the spatial resolution is 3.5 m. Fig. 3 shows the 3-D data cube and the distribution of targets of interest for the entire scene. The AVIRIS-1 used in our experiments is a subimage with 100 × 100 pixels. As shown in Fig. 4, there are three airplanes regarded as targets occupying 22, 22, and 20 pixels, respectively.

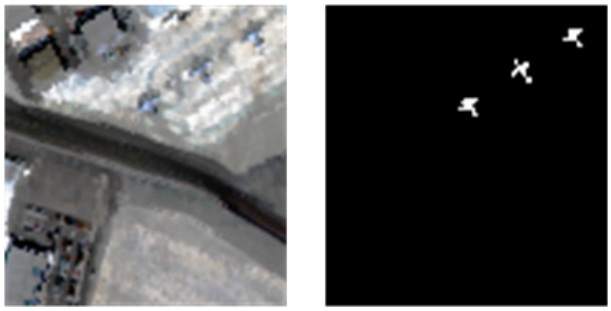
2) AVIRIS-2 Data Set: The second data set was also originally obtained by AVIRIS and consisted of 128 × 128 pixels selected from the entire scene, as shown in Fig. 3. In this subimage, the three airplanes to be detected contain 34, 42, and 44 pixels, respectively. There are 189 bands remaining after the removal of the poor-quality bands and the spatial

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(a) (b)

Fig. 3. (a) 3-D AVIRIS data cube. (b) Locations of targets in two subimages marked with different colors.



(a) (b)

Fig. 4. (a) True-color image of AVIRIS-1. (b) Ground-truth map.



(a) (b)

Fig. 5. (a) True-color image of AVIRIS-2. (b) Ground-truth map.

resolution is 3.5 m. The corresponding true-color image and the ground-truth map are illustrated in Fig. 5.

3) HyMap Data Set: The third data set was captured by the HyMap Airborne Hyperspectral Imaging Sensor on July 4, 2006. The full image covers an area of Cooke City, Montana, USA. It was published by the Rochester Institute of Technology (RIT) target detection project in New York, USA, and has become one of the most widely used benchmark data sets. The full image with a spatial resolution of 3 m contains 280 × 800 pixels and provides 126 spectral channels in the range of visible near-infrared (VNIR) to shortwave infrared (SWIR). The spectral resolutions is about 14 nm. We cropped a subimage with 100 × 300 pixels for experiments [63]. The 3-D hyperspectral cube of the entire scene is shown in Fig. 6; the dotted box marks the cropped area. There are seven types of targets of interest that vary in location, size, and shape,

TABLE II
TARGET INFORMATION FOR THE HYMAP DATA SET

Naming Convention	Type of Target	Description of the Occupied Pixels
F1	3m Red Cotton	3 × 3 Full : 1 Sub : 8
F2	3m Yellow Nylon	3 × 3 Full : 1 Sub : 8
F3a	1m & 2m Blue Cotton	3 × 3 Full : 1 Sub : 8
F3b		1 × 1 Sub : 1
F4a	1m & 2m Red Nylon	3 × 3 Full : 1 Sub : 8
F4b		1 × 1 Sub : 1
V1	1993 Chevy Blazer	1 × 1 Sub : 1
V2	1997 Toyota T100	1 × 1 Sub : 1
V3	1985 Subaru GL Wagon	1 × 1 Sub : 1

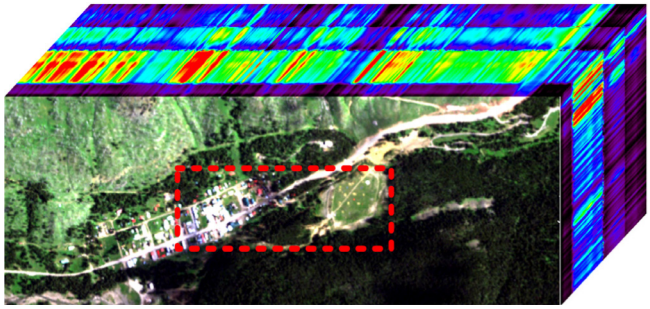


Fig. 6. 3-D HyMap data cube corresponding to the full image.

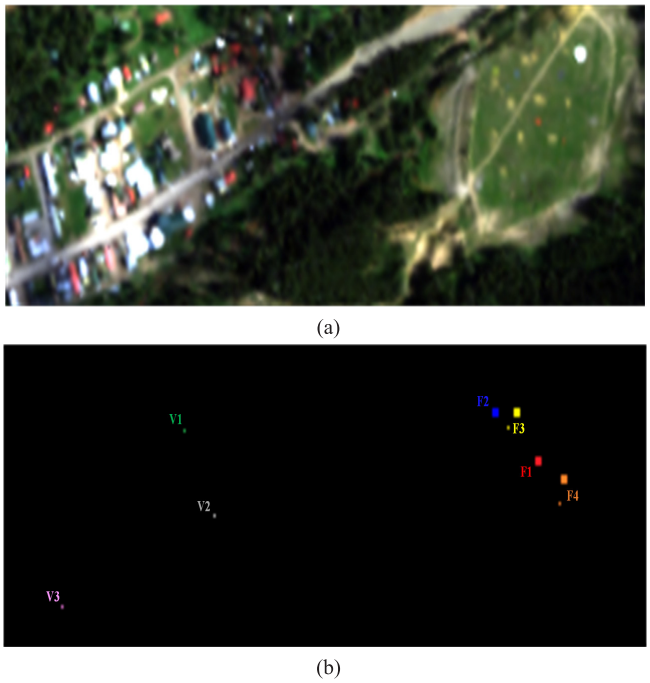
including fabric panels F1, F2, F3a, F3b, F4a, and F4b and vehicles V1, V2, and V3, as shown in Fig. 7. More information about the targets is listed in Table II.

To evaluate the performance of the detection algorithms fairly and reasonably, the quantitative indicators used in our experiments are summarized as follows.

1) Receiver Operating Characteristic Curve: The receiver operating characteristic (ROC) curve describes the relationship between the probability of detection and the FAR by setting different thresholds on the detection results [36]. The FAR is the abscissa of the ROC curve, indicating the probability of false alarms of the target under different thresholds. The ordinate is the probability of detection, which is the ratio of the number of true target pixels in the detection results to the total number of target pixels in the ground truth. The detection algorithm with ideal performance has low FAR and high probability of detection. Therefore, the more the ROC curve shifts to the upper left, the better the performance of the corresponding algorithm.

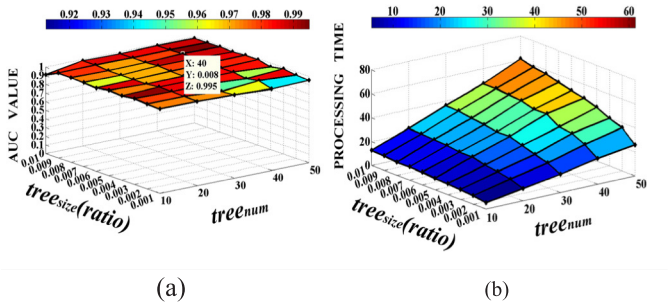
2) Area Under the ROC Curve (AUC): The area under the curve (AUC) represents the area under the ROC curve, which can be calculated by integrating the ROC [36]. The more the

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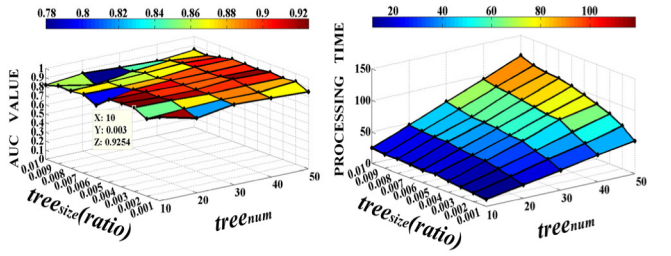
(a) (b)

Fig. 7. (a) True-color subimage of the HyMap data set. (b) Ground-truth map of seven types of targets marked with different colors.



(a) (b)

Fig. 8. Colored surface representing the relationship between $\text{tree}_{\text{size}}$(ratio), tree_{mm} combinations and the performance of TD-TSE on AVIRIS-1. (a) AUC values. (b) Processing time.



(a) (b)

Fig. 9. Colored surface representing the relationship between $\text{tree}_{\text{size}}$(ratio), tree_{mm} combinations and the performance of TD-TSE on AVIRIS-2. (a) AUC values. (b) Processing time.

ROC curve shifts to the upper left, the larger the value of AUC, suggesting the better detection performance.

3) False Alarm Rate: If a data set contains very few target pixels of interest, the ROC curves of different detectors can be fairly close together. In such cases, the FAR can be adopted as an evaluation indicator for comparison [37]. In units of pixels, FAR is obtained by calculating the ratio of the number of pixels misjudged as the target in the detection results to the total number of background pixels in the ground truth. Under the same probability of detection, detectors with lower FAR values perform better.

4) Separability Map: According to the information provided by the ground truth, the values corresponding to the locations of the target and background in the detection results can be divided into two sets. The separability map displays the statistical range of these two sets of values in the form of boxes with different colors [67]. The central mark of each box is the median, the edges are the 25th and 75th percentiles, and the whiskers extend to 0.5th and 99.5th percentiles. In this way, the separability between the target and the background is visualized, which also provides a reasonable basis for evaluating the validity of detectors.

B. Discussion on Parameters and Complexity Analysis

As demonstrated by Algorithm 2, there are two critical parameters involved in the input of the TD-TSE. In the construction of binary trees, $\text{tree}_{\text{size}}$ and tree_{num} denote the subsampling size and the number of trees, respectively. Considering both $\text{tree}_{\text{size}}$ and tree_{num} have an influence on the performance of the proposed detector, we perform experiments to find the optimal combinations of these two parameters for different data sets based on quantitative indicators. We evaluate the parameters simultaneously by setting ranges for $\text{tree}_{\text{size}}$ and tree_{mm}. As the three experimental data sets used have different sizes, it is more reasonable to use the ratio of the subsampling size to the total number of pixels for $\text{tree}_{\text{size}}$, and the range of ratios corresponding to $\text{tree}_{\text{size}}$ is set to [0.001, 0.002, 0.003, 0.004, 0.005, 0.006, 0.007, 0.008, 0.009, 0.01]. When experimenting with only one tree, in the tree construction stage through random subsampling, the utilization rate of pixels is not more than 1%, which cannot guarantee sufficient extraction of effective features of the target and the background in the hyperspectral data, resulting in quite unstable detection effects. Considering that the detection result using a single tree is not conducive to parameter analysis, so in our experiments, the range of tree_{mm} is set to [10, 20, 30, 40, 50]. The same parameter combinations of $\text{tree}_{\text{size}}$ and tree_{mm} are used for all three data sets.

For the AVIRIS-1 data set, Table III gives the processing time and AUC values of the proposed TD-TSE under different $\text{tree}_{\text{size}}$(ratio), tree_{mm}. According to Table III, Fig. 8 illustrates the relationship between the parameters and the detector performance more intuitively. Under 50 parameter combinations, the AUC varies from 0.9126 to 0.9950. When $\text{tree}_{\text{size}}$(ratio) $= 0.008$ and tree_{mm}, the TD-TSE's AUC value reaches the maximum. As shown in Fig. 8(a), the AUC value of the proposed method changes smoothly with $\text{tree}_{\text{size}}$(ratio) and tree_{mm} in general. For the AVIRIS-1 containing three relatively large and similar-sized airplane targets, the detection effect of the TD-TSE is not particularly sensitive to $\text{tree}_{\text{size}}$(ratio)

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TABLE III
PROCESSING TIME AND AUC VALUES OF THE TD-TSE WITH DIFFERENT PARAMETER COMBINATIONS ON AVIRIS-1

$tree_{num}$	10		20		30		40		50	
$tree_{size}$(ratio)	time(s)	AUC	time(s)	AUC	time(s)	AUC	time(s)	AUC	time(s)	AUC
0.001	5.4799	0.9743	10.3123	0.9729	15.1979	0.9620	21.4290	0.9415	25.7626	0.9515
0.002	7.2834	0.9795	14.0588	0.9828	20.5493	0.9500	29.1140	0.9852	36.1967	0.9619
0.003	8.3812	0.9933	16.2416	0.9851	23.8694	0.9825	34.2437	0.9571	39.8097	0.9824
0.004	9.2727	0.9738	17.6523	0.9859	27.0259	0.9846	35.6393	0.9827	44.9645	0.9556
0.005	9.7380	0.9861	20.6574	0.9693	30.6375	0.9737	42.6196	0.9810	48.6242	0.9885
0.006	10.8167	0.9612	20.5430	0.9695	30.3750	0.9786	42.1173	0.9847	50.8483	0.9911
0.007	11.0323	0.9737	21.9657	0.9693	32.3387	0.9890	45.4050	0.9874	53.7024	0.9768
0.008	10.9482	0.9804	22.7769	0.9713	33.7527	0.9788	46.3367	**0.9950**	56.7168	0.9841
0.009	12.3353	0.9812	23.3582	0.9866	35.6664	0.9855	47.3152	0.9877	58.2675	0.9801
0.010	12.7846	0.9126	24.0362	0.9853	36.1831	0.9794	47.3712	0.9845	61.3220	0.9826

and $tree_{mm}$, which illustrates the robustness of the proposed method. Fig. 8(b) visualizes the relationship between the processing time and the parameters. It can be seen that the processing time of TD-TSE is close to being linearly related to the number of trees $tree_{mm}$. As the subsampling ratio $tree_{size}$(ratio) becomes larger, the time required by TD-TSE increases roughly linearly but more slowly. This is because in most cases of binary tree construction, the tree does not fully grow and the processing time is less than for the worst case.

For the AVIRIS-2 data set, Table IV and its corresponding colored surface in Fig. 9 provide the experimental results of the parameters $tree_{size}$(ratio) and $tree_{mm}$ for the proposed TD-TSE. AUC varies from 0.7794 to 0.9254, and $tree_{size}(ratio) = 0.003$, $tree_{mm} = 10$ is the optimal parameter combination for the detection effect. When $tree_{mm}$ is less then 30, the AUC fluctuates obviously in Fig. 9(a). Unlike AVIRIS-1, the sizes of three targets in AVIRIS-2 are quite different, which means that for the subsampling with less times, valid features of target and background cannot be fully extracted in the construction of binary trees. The fluctuation of AUC in Fig. 9(a) can thus be explained by the distribution characteristics of the targets in AVIRIS-2. It can be seen that as $tree_{size}$(ratio) and $tree_{mm}$ become larger, the AUC values grow and tend to converge. For HSIs with significantly different target sizes, the stability of the TD-TSE can be improved by increasing the number of trees. Apart from these, Fig. 9(b) further illustrates the linear time complexity of the proposed detector.

For the HyMap data set, FAR is adopted as the evaluation indicator. It is intended that several classic hyperspectral detectors will be used in the subsequent experiments for algorithm comparison. Based on the information in Table II, each target in the HyMap data occupies very few pixels, which will cause the ROC curves of different detectors to be fairly close together. Given this, FAR, which is a quantitative indicator, would be able to distinguish the performance of different detectors more clearly for comparisons. The lower the FAR, the lower the probability that the detector misjudges the background as the target, the better the detection performance. Accordingly, the optimal $tree_{size}$(ratio) and $tree_{mm}$ for seven types of targets under 50 combinations are listed in Table V. The optimal parameter settings of AVIRIS-1 and AVIRIS-2 discussed earlier are also provided for the comparative experiments in Section III-C.

The aforementioned ranges set for $tree_{size}$(ratio) and $tree_{mm}$ are 0.001–0.01 and 10–50, respectively. For experimental data sets in real scenes with different sizes of targets of interest, the subsampling for each tree using only 0.2%–1.0% of the total number of pixels and the construction of about 10–40 trees can achieve excellent detection performance. Compared with the spatial size of the hyperspectral data sets used in experiments, the proposed method does not have a high utilization rate of pixels in the tree construction stage. Based on the concept of ensemble learning, the parallel measurements of the distance on tree are implemented by a certain number of small-scale subsampling to reduce the generalization error generated in the tree construction. Unlike the classical forest-based methods in ensemble learning, the number of trees required by the proposed method is much smaller. As the optimal parameter settings on different data sets shown in Table V, the TD-TSE performs well with small $tree_{size}$(ratio) and $tree_{mm}$, indicating low requirements for memory and processing time. The computational efficiency demonstrated in experiments proves that our detector is practical for HSIs in real scenes.

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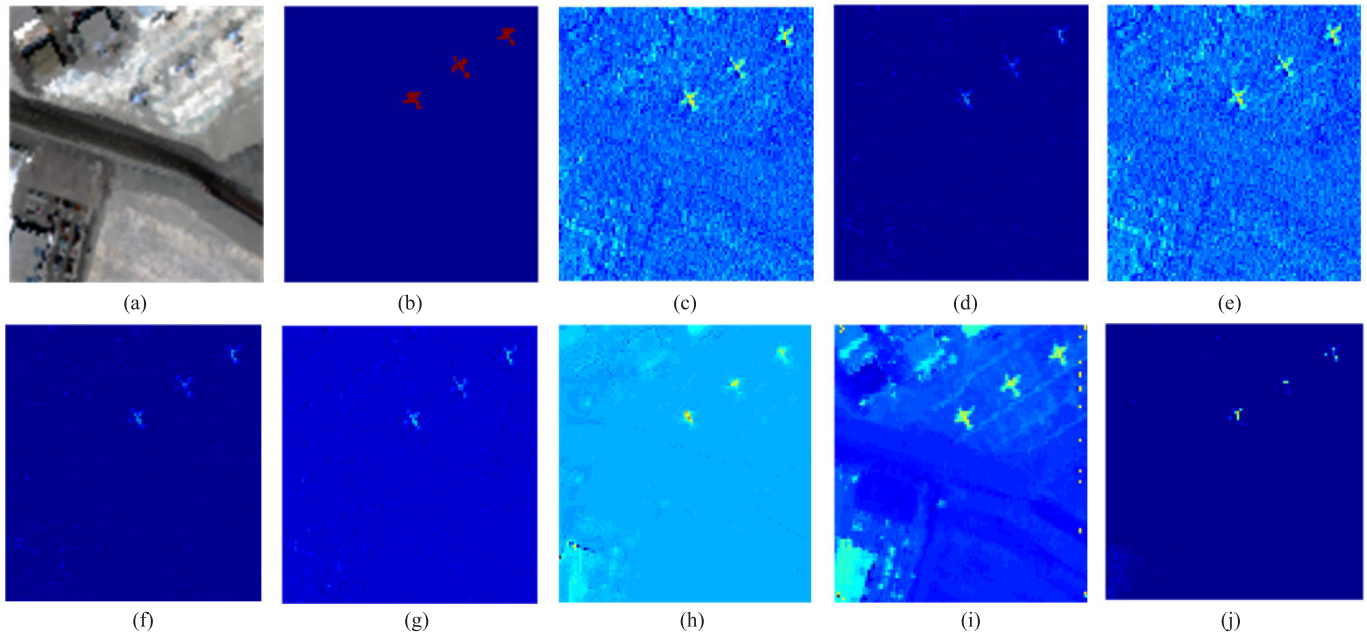


Fig. 10. Colored maps for comparison of detection algorithms on the AVIRIS-1 data set. (a) True color. (b) Ground truth. (c) MF. (d) GLRT. (e) CEM. (f) ACE. (g) S-ACE. (h) SRD. (i) CSCR. (j) TD-TSE.

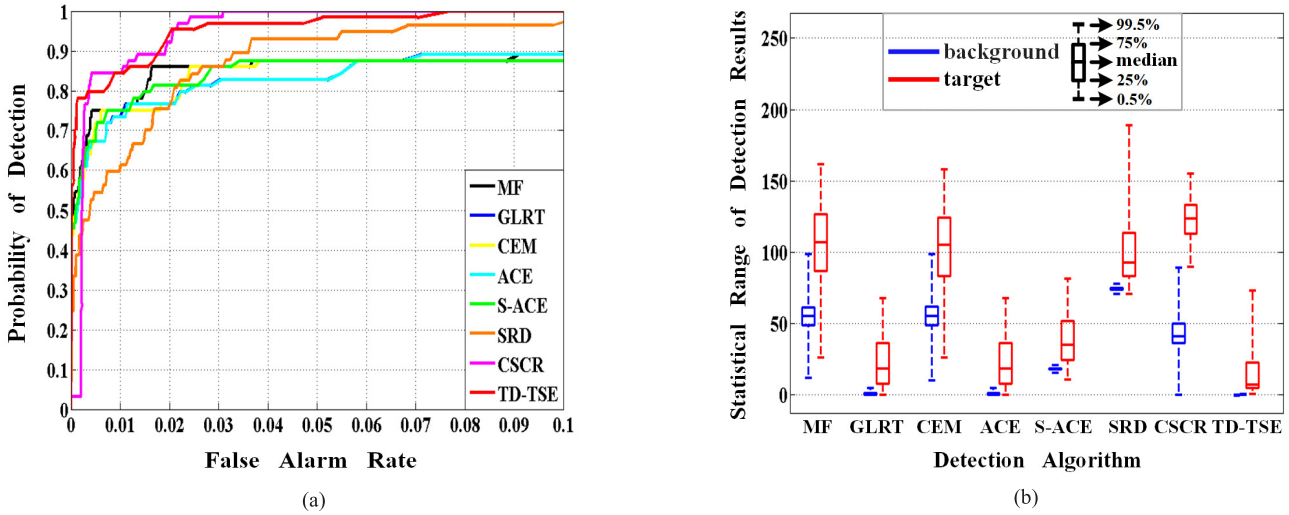


Fig. 11. ROC curves and separability map for different detection algorithms on the AVIRIS-1 data set. (a) ROC curves. (b) Separability map.

C. Comparison of Detection Performance

To make a fair evaluation and fully demonstrate the effectiveness of the proposed TD-TSE, a selection of the most classic and widely used detection algorithms is applied to the experimental data sets for comparisons. The algorithms designed to compare with the proposed method on three experimental data sets include the following: 1) matched filter (MF) [4]; 2) GLRT; 3) CEM; 4) ACE; 5) signed ACE (S-ACE) [68]; 6) SRD; and 7) CSCR. The SRD and CSCR work with a sliding dual-window strategy, and the dual window is used to select pixels for building the background dictionary adaptively. Given that the target sizes in three experimental data sets are different, we set the appropriate sizes of dual window to achieve the excellent detection performance for the SRD and CSCR. For AVIRIS-1, the sizes of the outer and inner windows are 15×15 and 5×5, respectively. For AVIRIS-2, the sizes of the dual window are 17×17 and 7×7, respectively. For the HyMap, the window sizes are set to 13×13 and 3×3. In each data set, all the aforementioned detectors use the same given target spectrum as the prior information, so that the comparison can be made fairly to verify the innovation and validity of the proposed method. In subsequent experiments, the parameters of TD-TSE on all three data sets are set according to the optimal combinations shown in Table V.

For the AVIRIS-1 data set, the 2-D detection results of the HSI are linearly stretched to the range 0–255. Fig. 10 shows the colored maps derived in this way for comparison. The corresponding ROC curves and separability map are given in Fig. 11. According to the ground truth, it can be seen

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TABLE IV
PROCESSING TIME AND AUC VALUES OF THE TD-TSE WITH DIFFERENT PARAMETER COMBINATIONS ON AVIRIS-2

$tree_{num}$	10		20		30		40		50	
$tree_{size}$(ratio)	time(s)	AUC	time(s)	AUC	time(s)	AUC	time(s)	AUC	time(s)	AUC
0.001	11.3849	0.9207	21.2115	0.8203	31.1268	0.8859	42.3839	0.8691	51.3026	0.8505
0.002	13.7811	0.8500	27.1702	0.8721	40.8834	0.8875	52.8306	0.8800	67.6659	0.9127
0.003	15.9416	**0.9254**	31.3351	0.9177	46.7391	0.9003	63.4965	0.9008	79.0871	0.8997
0.004	17.8999	0.9215	36.3351	0.8992	54.9634	0.8933	75.5669	0.9229	87.1073	0.9133
0.005	21.3654	0.9122	37.8060	0.9078	56.2043	0.8935	76.2367	0.9010	94.3805	0.8918
0.006	19.6048	0.7967	39.9420	0.8814	59.5159	0.8937	80.1219	0.8944	100.174	0.9027
0.007	21.0257	0.8723	43.7016	0.8636	61.1114	0.8688	83.6248	0.9051	105.149	0.9054
0.008	22.3023	0.8553	44.0190	0.8587	62.3325	0.8737	87.6278	0.8960	106.548	0.8742
0.009	22.0689	0.8582	45.0475	0.7794	66.9139	0.8435	89.1873	0.8728	110.459	0.8438
0.010	24.5575	0.8153	47.0922	0.8594	71.4324	0.8350	93.6056	0.8195	118.587	0.8568

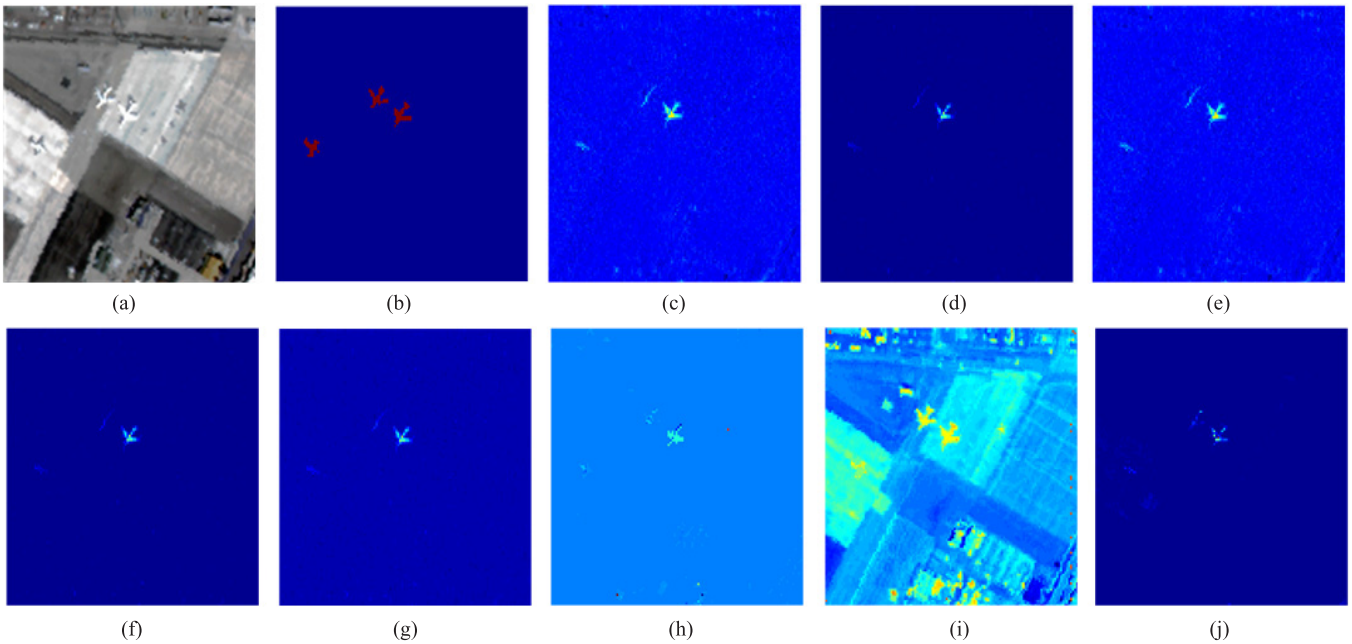


Fig. 12. Colored maps for comparison of detection algorithms on the AVIRIS-2 data set. (a) True color. (b) Ground truth. (c) MF. (d) GLRT. (e) CEM. (f) ACE. (g) S-ACE. (h) SRD. (i) CSCR. (j) TD-TSE.

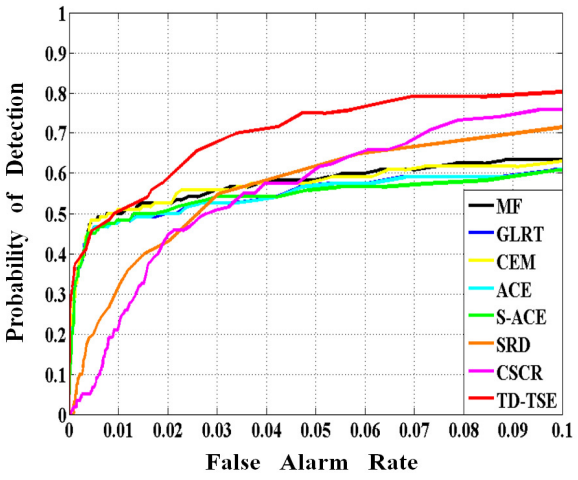
that all the detectors are able to highlight the targets to some extent. For MF, CEM, SRD, and CSCR, the spatial information of three airplanes in the detection colored maps is substantially completely preserved, so we can basically see the original shape of the targets. However, the upper and lower left buildings in the urban scene are also highlighted, which leads to the possibility that some prominent background pixels are judged as targets. Besides, because the dual-window detection mode requires boundary expansion, CSCR also generates some obvious false alarms at the edge of the image. In short, the aforementioned MF, CEM, SRD, and CSCR cause the relatively significant background interference while highlighting the target. In contrast, GLRT, ACE, S-ACE, and the proposed TD-TSE all perform well at reducing false

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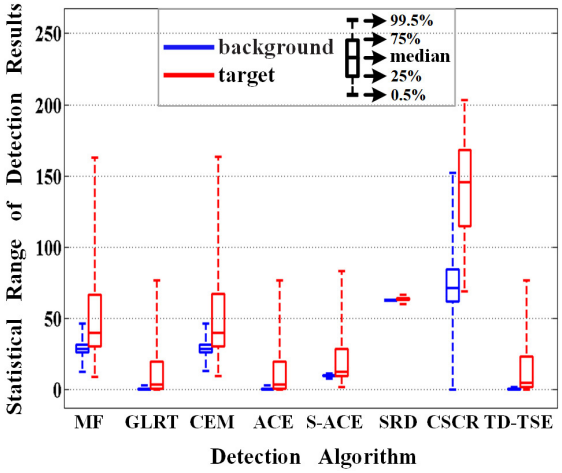
TABLE V
OPTIMAL PARAMETER SETTINGS FOR THE TD-TSE ON EXPERIMENTAL DATA SETS

Dataset		Evaluation		Optimal Parameter Setting	
		Indicator | Result		$tree_{size}(ratio)$	$tree_{num}$
AVIRIS-1		AUC	0.9950	0.008	40
AVIRIS-2			0.9254	0.003	10
HyMap	F1	FAR	6.6687e-05	0.002	20
	F2		6.6687e-05	0.005	10
	F3a		0	0.003	10
	F3b		1.0000e-04	0.006	20
	F4a		2.0006e-04	0.003	30
	F4b		1.0000e-04	0.01	20
	V1		6.6669e-05	0.004	10
	V2		0	0.005	20
	V3		3.3334e-05	0.01	30

alarms, but it can be found that the detection values of TD-TSE at background and target locations are the most easily separable. Comprehensively considering highlighting the target and suppressing the background, the proposed method is superior to other algorithms in the visual comparison on AVIRIS-1. Fig. 11(a) shows the ROC evaluation of all detectors. The inferiority of the algorithms MF, GLRT, CEM, ACE, and S-ACE in processing background or target signals causes their ROC curves to intersect each other while being generally close. The SRD performs poorly at the beginning; after FAR = 0.03, its probability of detection is higher than that of the previous five traditional detectors. The ROC curves of the CSCR and the proposed TD-TSE are notable for being closest to the upper left, indicating that these two methods detect the objects of interest with higher confidence. Fig. 11(b) further visualizes the separability of the target and the background for all the detection results. Counting the whiskers, the intersecting range between two boxes of MF and CEM is large, indicating the poor separation effect for the target and background. The detection results of GLRT, ACE, S-ACE, SRD, CSCR, and TD-TSE provide better separability between the two sets of values. Among them, the amount of intersection between the two boxes for CSCR and TD-TSE is minimal, which represents the ideal performance in separating the target from the background. It is worth noting that the blue box representing the background in the CSCR detection results occupies a certain range, while the blue box of the TD-TSE is at the lowest position. Although the ROC curves of these two methods cannot tell who has better detection performance, this difference in the separability map shows the absolute advantage of the proposed TD-TSE in suppressing background interference.



(a)



(b)

Fig. 13. ROC curves and separability map for different detection algorithms on the AVIRIS-2 data set. (a) ROC curves. (b) Separability map.

For the AVIRIS-2 data set, preliminary conclusions can be drawn from the detection colored maps shown in Fig. 12. Although MF, CEM, SRD, and CSCR can detect the airplanes of interest in a specific scene, the suppression of background interference is significantly inferior to that of GLRT, ACE, S-ACE, and TD-TSE. At the bottom of the colored maps of MF and CEM, there are a few highlighted background pixels. The overall detection output values for SRD are at a high level, illustrating the poor ability to separate the target from the background. As for CSCR, the contours of land cover in the colored map is relatively clear, which also causes a certain degree of false alarms. The visual difference between the 2-D detection results of GLRT, ACE, S-ACE, and TD-TSE is not quite obvious, so the ROC curves in Fig. 13(a) can be used for a quantitative comparison. When FAR is greater than 0.01, the ROC curve of the proposed method performs significantly better than those of the other methods. It can be deduced that with the increase in the segmentation threshold, the TD-TSE can accurately identify the target pixels and maintain

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TABLE VI
FAR EVALUATION RESULTS FOR COMPARISON OF DETECTION ALGORITHMS ON THE HYMAP DATA SET

	MF	GLRT	CEM	ACE	S-ACE	SRD	CSCR	TD-TSE
F1	2.0006e-04	**6.6687e-05**	2.3340e-04	1.0003e-04	3.3343e-04	6.3352e-04	1.2004e-03	**6.6687e-05**
F2	1.3337e-04	**3.3378e-05**	1.6672e-04	1.3337e-04	5.3349e-04	7.3355e-04	3.3343e-04	6.6687e-05
F3a	7.7357e-03	**0**	1.0203e-02	3.3334e-05	4.7348e-03	1.0036e-02	1.6005e-03	**0**
F3b	8.0669e-03	2.8334e-04	1.0400e-02	3.0001e-04	6.2669e-03	3.9835e-02	3.2668e-03	**1.0000e-04**
F4a	3.0009e-04	**0**	3.0009e-04	**0**	1.9673e-03	8.0024e-04	3.3343e-04	2.0006e-04
F4b	4.6668e-04	1.6667e-04	9.6670e-04	2.0001e-04	3.0334e-03	1.7334e-03	1.0334e-03	**1.0000e-04**
V1	5.9935e-02	1.0000e-04	6.8302e-02	1.6689e-04	3.9268e-02	2.3334e-04	2.0001e-04	**6.6669e-05**
V2	1.1900e-02	3.3334e-05	6.0335e-02	3.3334e-05	4.3501e-02	1.1334e-03	1.6667e-04	**0**
V3	3.3534e-02	1.3334e-04	5.1368e-02	1.6689e-04	4.9468e-02	1.0867e-02	8.6670e-04	**3.3334e-05**

TABLE VII
PROCESSING TIME FOR COMPARISON OF DETECTION ALGORITHMS ON EXPERIMENTAL DATA SETS

Dataset		**Processing Time of the Detection Algorithms(s)**							
		MF	GLRT	CEM	ACE	S-ACE	SRD	CSCR	TD-TSE
AVIRIS-1		0.7490	1.1459	0.3182	0.7509	1.2791	690.405	41.2569	46.3367
AVIRIS-2		0.9833	2.0926	0.5106	1.2640	1.7940	1011.20	95.0005	15.9416
HyMap	F1	1.1948	2.1930	0.7621	1.5479	2.3373	914.300	81.1634	69.1920
	F2	0.9228	2.3275	0.7300	1.6659	2.2494	938.209	94.2491	80.2302
	F3a	1.0235	2.3250	0.8438	1.8320	2.4262	916.473	83.0118	46.9271
	F3b	1.0006	2.4503	0.7979	1.7860	2.3661	917.974	77.4531	96.4143
	F4a	1.0204	2.3857	0.7744	1.7370	2.2018	914.589	78.9788	135.978
	F4b	1.0021	2.2966	0.7161	1.8051	2.2075	940.278	79.4154	146.149
	V1	1.0300	2.4979	0.7892	1.7540	2.2919	938.281	78.2779	56.1506
	V2	1.0352	2.5602	0.7606	1.7302	2.2822	950.879	82.9346	107.035
	V3	1.1106	2.5172	0.7743	1.7125	2.3420	935.437	79.7394	211.497
Processing Time (sum)		11.0723	24.7919	7.7772	17.5855	23.7775	10068.0	871.481	1011.85

a high detection rate, which benefits from the output of the detector designed based on the distance on tree. In particular, the ROC curves of the SRD and CSCR show the same overall trend. With FAR = 0.05 as the approximate cut-off point, the SRD and CSCR perform worse than other algorithms before, but their detection performance is only inferior to TD-TSE after that. In the separability map shown in Fig. 13(b), for MF, CEM, SRD, and CSCR, 0.5%–99.5% of the detection values at background locations all intersect with the target boxes, and the overlaps account for a large proportion of the targets. In contrast, GLRT, ACE, S-ACE, and TD-TSE have relatively small intersection areas between two boxes for the target statistics. Among these four detectors, the position of the background box for S-ACE is higher, which means that there is background interference under certain thresholds and false alarms are generated. The nuances of the other three detectors GLRT, ACE, and TD-TSE are reflected in the overall ranges of detection values for the backgrounds. Careful observation reveals that the proposed TD-TSE has the strongest inhibitory effect on backgrounds.

For the HyMap data set, since there are seven types of targets occupying only a few pixels, FAR is adopted and the evaluation results are listed in Table VI for comparison. For each target, the lowest detection FAR is marked in bold.

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TABLE VIII
EVALUATION RESULTS FOR COMPARISON OF DETECTION ALGORITHMS ON EXPERIMENTAL DATA SETS

Dataset	AVIRIS-1	AVIRIS-2	HyMap
Indicator	AUC	AUC	FAR (sum)
MF	0.9340	0.7827	1.2227e-01
GLRT	0.9379	0.8036	8.1675e-04
CEM	0.9298	0.7826	2.0227e-01
ACE	0.9379	0.8034	1.1339e-03
S-ACE	0.9347	0.7813	1.4911e-01
SRD	0.9664	0.8533	6.6005e-02
CSCR	0.9949	0.9146	9.0013e-03
TD-TSE	**0.9950**	**0.9254**	**6.3344e-04**

Except for F2 and F4a, the results of TD-TSE have been marked seven times among all targets at nine locations, and its FAR value of F2 is on the same order of magnitude 10^{-5} as the lowest FAR. Compared with other algorithms, the proposed method has the lower FAR on the whole level, indicating the superior detection ability with high confidence. It is worth noting that the information about the targets given in Table II shows that the nine pixels for each of F1, F2, F3a, and F4a contain eight subpixels, and the other targets all exist in the form of subpixels. The poor separability of the targets and backgrounds is the technical difficulty for subpixel target detection. Based on the numerical difference between various land cover spectra, the segmentation characteristic of tree structure ensures that the subpixel target and background are separated at the less correlated spectral bands. Furthermore, the distance on tree measures the separation between the test pixel and the given target spectrum effectively. The theoretical strategy of the proposed method reasonably explains the outstanding performance of TD-TSE in the subpixel target detection experiments on HyMap.

Table VII gives the processing time on all types of targets for the proposed method and other comparison algorithms. In particular, for F3a and F3b, although the same given spectrum is used as the prior information, these two target locations correspond to different optimal parameter settings for TD-TSE. To be fair, other comparison algorithms are also executed twice for the experiment on F3, the same for F4. The total processing time of the detectors on all targets is given at the end of Table VII. Table VIII summarizes the quantitative comparison results of three experimental data sets. For each detector, in the case of the HyMap, the sum of FARs on the targets at nine locations is provided for overall evaluation. The best evaluation results and the corresponding algorithms are marked in bold. Considering both the processing time and evaluation results, we can see that the traditional MF, GLRT, CEM, ACE, and S-ACE only take very little time, but their AUC values on AVIRIS-1 and AVIRIS-2 are smaller than those of SRD, CSCR, and TD-TSE. The three algorithms based on machine learning are superior in terms of detection effect at the cost of processing time. The SRD consumes large amounts of time because the orthogonal matching pursuit (OMP) is adopted to recover the sparse representation [47], and the running time of TD-TSE and CSCR is only one-tenth or less of the time required by SRD. The experimental results of HyMap reveal the disadvantages of two representation-based detectors, SRD and CSCR, in terms of suppressing false alarms. Among the comparison algorithms, the performance of GLRT is quite stable, but TD-TSE still takes the leading position with a slight advantage. Although it takes more time to run, the detection effect of TD-TSE is significantly better than that of the traditional algorithms, and its time consumption is considerable compared with the representation-based methods. Generally, the proposed method exhibits ideal comprehensive performance and is highly practical for hyperspectral target detection in different real scenes.

IV. CONCLUSION

In this article, we propose a TD-TSE to overcome the limitations of the traditional model-driven detection methods. The binary tree is constructed by random subsampling and used for image segmentation to improve the separability of the target and the background. Based on the tree structure, the distance on tree between the test pixel and the given target spectrum is skillfully calculated via binary encoding for the detection output. To reduce the generalization error generated by random subsampling, a collection of independent trees are constructed for the parallel distance measurements, which further improves the robustness of the proposed method. Extensive experimental results of different data sets prove that benefiting from the effective measurement, the distance on tree, TD-TSE with linear complexity can achieve the ideal performance by carrying out a certain number of small-scale subsampling. This makes the TD-TSE more practical for real hyperspectral data. Comparison with other algorithms demonstrates the superiority of the proposed method over state-of-the-art.

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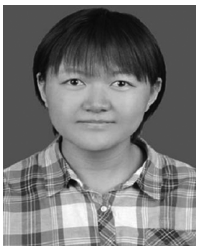
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